

Boyuan Zhang

✉ boyuan.zhang@cs.jhu.edu  gary-boyuan-zhang.github.io  gary-boyuan-zhang

Education

Johns Hopkins University

Ph.D. in Computer Science

Baltimore, MD

Aug. 2026 – Present

Stanford University

M.S. in Statistics

Stanford, CA

Sep. 2024 – Jun. 2026

New York University

B.A. in Computer and Data Science

New York, NY

Jan. 2021 – Dec. 2023

Research Interests

Causal inference, statistical inference, machine learning, and AI evaluation, with applications in healthcare.

Preprints

Treatment Response as a Latent Variable

Christopher Tosh, *Boyuan Zhang*, Wesley Tansey

[arXiv:2502.08776](https://arxiv.org/abs/2502.08776) 

In Submission

at JRSS-B

Research Experience

Stanford University, Department of Statistics

Advisor: Wing Hung Wong

Stanford, CA

June 2025 – Present

- Developed a generative representation-learning framework for causal mediation analysis with high-dimensional confounding, enabling estimation of direct and indirect effects across binary, continuous, and multivariate treatment and mediator settings.

Stanford Law School, Deborah L. Rhode Center on the Legal Profession

Advisor: Aviv Caspi

Stanford, CA

June 2025 – Present

- Developed a large-scale mobility-analysis pipeline using device-level location data to identify household moves around eviction filings and characterize post-eviction displacement patterns in Los Angeles County.

Stanford University, Department of Management Science and Engineering

Advisor: Itai Ashlagi

Stanford, CA

Mar. 2025 – Jun. 2026

- Analyzed kidney allocation policies and transplant-center behavior to study efficiency-equity trade-offs, expedited organ offers, and strategic incentives in organ acceptance and patient listing decisions.

Memorial Sloan Kettering Cancer Center

Advisor: Wesley Tansey

New York, NY

Jun. 2023 – May 2024

- Developed a nonparametric procedure for testing individual treatment effects under confounding and non-compliance, controlling false discovery rates while identifying treatment-response biomarkers in large-scale cancer drug data.

New York University, Courant Institute of Mathematical Sciences

Advisor: Daniel Neill

New York, NY

Feb. 2023 – May 2024

- Developed subset-scanning methods for automated difference-in-differences discovery, jointly identifying heterogeneous treatment discontinuities and statistically valid control subgroups in high-dimensional observational data.

Carnegie Mellon University, Department of Statistics and Data Science

Advisors: Ronald Yurko, Konstantinos Pelechrinis

Pittsburgh, PA

Jun. 2022 – Jul. 2022

- Developed a Bayesian regularized adjusted plus-minus model for soccer player evaluation, incorporating box-score-informed priors to improve predictive performance and interpretability over existing approaches.

Teaching Experience

New York University, Center for Data Science & Courant Institute

Teaching Assistant

New York, NY

Sep. 2022 – Dec. 2023

DS-UA 301 Advanced Techniques in Machine Learning and Deep Learning

Fall '23

DS-UA 201 Causal Inference

Summer '23

CSCI-UA 2 Intro to Computer Programming

Fall '22 '23, Spring '23

Technical Skills

Python, R, SQL, Java, C/C++, L^AT_EX, Git, Slurm/HPC